Solving Counting Problems

A counting problem asks you to count the possible ways that something could happen. In some problems, you will need to use operations to find the number of possibilities. However, in many simple counting problems, all you need to do is organize your work and count carefully.

Example 1

Hide

The Parkers are taking a photograph of their four children—Alan, Bianca, Callie, and Dennis. They plan to stand in a line. How many different ways can the children line up?

To find the answer, you can use the first letters of their names to represent the children and write all the possible orders.

Step 1

Write the orders that begin with Alan (A).

ABCD
ABDC — Two start with AB.

ACBD
ACDB — Two start with AC.

ADBC
ADCB — Two start with AD.

Step 2

These begin with Bianca (B).

BACD
BADC

BCAD
BCDA

BDAC
BDCA

Step 3

These begin with Callie (C).

CABD
CADB

CBAD
CBDA

CDAB
CDBA

Step 4

These begin with Dennis (D).

DABC
DACB

DBAC
DBCA

DCAB
DCBA

Count the orders. There are **24 possible ways** in all.

If you carefully organize your work, you can often see a pattern that will help you count the possibilities. Once you know that there are 6 ways that start with Alan, you can conclude that there will be 6 ways for each child: $6 \times 4 = 24$ possible ways.

A diagram is another way to organize your work.

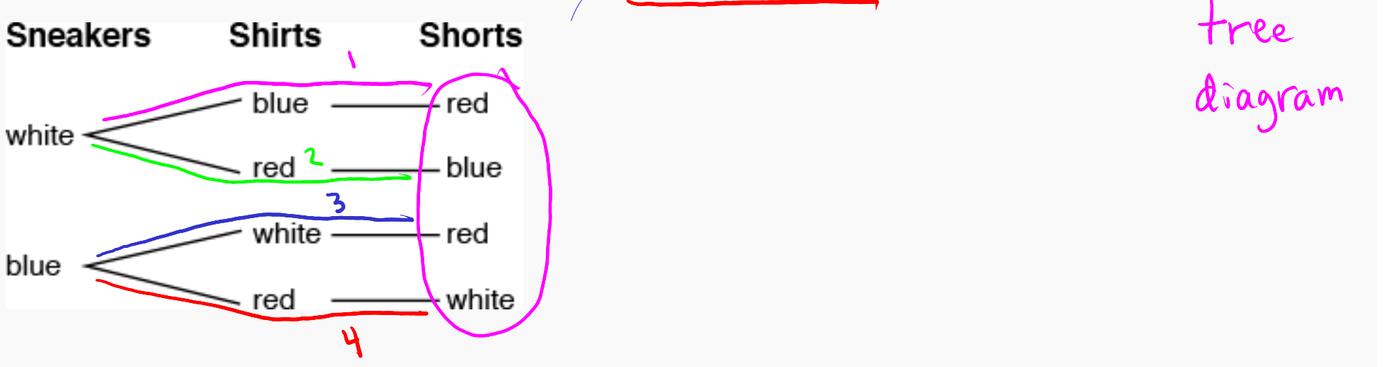
Example 2

Hide

Emily wants to wear red, white, and blue for a holiday. She can choose from two pairs of sneakers—blue or white. She has three shirts—red, white, or blue—and three pairs of shorts—red, white, or blue. She wants to wear one red item, one white item, and one blue item. How many outfits can she make?

This problem sounds more complicated than it really is. One way to solve the problem is to draw a diagram with columns for each type of clothing. Then draw lines to connect colors to make outfits.

As you can see from the diagram, Emily can make **four possible outfits**.



In the problems that follow, try different strategies. You can make an organized list, create a diagram, or even draw a picture. Good problem solvers look for meaningful ways to organize the information in the problem. As they organize, they look for patterns that will help them find the answer.

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Lesson 4: Counting Problems

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